RODRIGO GOMES DE ARAÚJO

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Education

Master's — **Sep 2025 – Present (Jul 2027)**
Computer Science — *University of Porto (FCUP)*

Bachelor's — **Sep 2022 – Jul 2025**
Computer Science and Engineering — *University of Porto (FEUP)*

Experience

Co-founder & Lead Engineer — **Jul 2026 – Present**
Tinker (Devtools Startup) — *Rust, Git*

- Co-founded and lead a 3-person startup. Primary author of the codebase and owner of the architecture. Tinker merges version control, project management and CI into one system.
- Designed the trust model. Signed approvals bound to the exact action and a tamper-evident history linking every change in production to its spec, verification and approval.

Projects

Portuguese Elections Data Platform
Advanced Topics in Databases · 18/20 — *Python, PostgreSQL, PostGIS, SQL*

- ETL pipeline in Python normalizing 10 years of official results (8 elections) from inconsistent government spreadsheets. Layout detection, entity resolution of parties and idempotent loads with SHA-256 provenance.
- Every load is reconciled against official totals (votes, turnout, seats) and invariant checks must return zero. Triggers reject bad writes and D'Hondt seats are recomputed in SQL as a cross-check.
- Star-schema warehouse with CUBE/ROLLUP, window functions and drill-down to 3,000+ parishes.

Cloud-Native Hospital Backend
Big Data and Cloud Computing — *Python, Flask, Google Cloud, Artillery*

- REST API on Google Cloud for patient records, admissions, media and patient–doctor questions, with access control enforced server-side.
- Storage split by access pattern. Structured records in Firestore, binary files as blobs in Cloud Storage with only their metadata indexed and precomputed aggregates in dedicated cache collections.
- Heavy aggregations moved to scheduled serverless functions with cached views, so derived endpoints respond as fast as plain reads.

UFC Fight Prediction
Artificial Intelligence · 20/20 — *Python*

- Statistical model to predict the outcome of UFC fights from fighter data.

Flight Management System
Algorithms and Data Structures — *C++17, Graph Algorithms*

- Global airline network modelled as a directed graph, loaded from CSV. BFS routing with the fewest stops and search filtered by airline with minimal airline changes.
- Tarjan's algorithm to find the airports critical to the network's connectivity and nearest-airport search (Haversine).

Distributed Protocols
Distributed Systems · 19/20 — *Rust, Tokio, gRPC, Docker*

- Totally ordered multicast with Lamport clocks, token ring mutual exclusion with automatic ring healing and push-pull gossip with anti-entropy.

Research

CaST (Cryptographically-aware Session Types) — **Jul 2026 – Present (Jul 2027)**
Master's Thesis · Advisors: Pedro Ângelo, Bernardo Portela — *FreeST, Haskell, Type Systems*

- Extending session types so that misuse of cryptographic resources in protocols is rejected at compile-time, with a soundness proof and integration into the FreeST compiler.

Awards

4th Place National · Canguru Matemático (27,505 participants)

1st Place National Robotics Championship · RoboCup 2016 (Leipzig) — **May 2016**

Technical Skills

Languages: Python, C++, SQL, Rust, C, Java, Haskell, OCaml
Data & Tools: PostgreSQL, MongoDB, Firestore, pandas, NumPy, scikit-learn, FastAPI, Flask, Google Cloud, Docker, Git
Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)